

WORKFLOW & HIGH-LEVEL DESIGN

EduOS

Tutoring operations platform — not an LMS. Multi-tenant MVP architecture, roles, nine layers, and core workflows.

Product: EduOS (tutor-hub) · Version: local-first demo · 2026

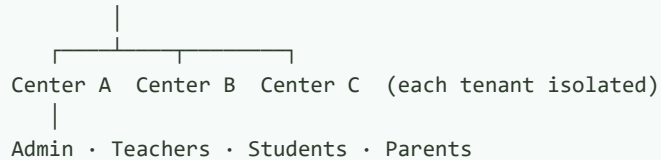
1. What EduOS Is

EduOS is the **operating system for tuition centers**. It handles everything around teaching:

- Acquiring and enrolling students (CRM)
- Running batches, schedules, attendance, and tests
- Messaging parents and staff
- Tracking student/tutor success signals
- Business intelligence and decisions
- Connecting external tools (Zoom, WhatsApp, calendars)

Teaching itself happens on the tutor's own platforms (Zoom, Google Meet, Khan Academy, etc.). EduOS coordinates logistics and visibility.

PLATFORM OWNER (EduOS) – onboard centers · monitor · support view



2. Design Principles

Principle	Meaning
Ops-first	CRM, schedule, attendance, comms — not course hosting
Multi-tenant	Each center's data isolated by <code>centerId</code>
Role-based portals	Same app, different nav and permissions per role
Decisions, not dashboards	BI suggests what to do next
Local MVP	Data in browser localStorage; run via <code>start.bat</code>

Deferred: Payments / billing (Layer 10 removed from roadmap).

3. Roles & Portals

Each user logs in through a portal tile. Session stored in `tutorhub_session`.

Role	Who	Default landing	Data scope
Platform owner	EduOS operator	Platform dashboard	All centers; support view into one center
Center admin	Academy owner	Command dashboard	Full center workspace
Teacher	Tutor	Teacher Today	Own batches & students
Student	Learner	Student home	Own profile, schedule, homework
Parent	Guardian	Parent dashboard	Linked child(ren) only

PLATFORM OWNER SUPPORT VIEW

All centers → open a center → Support view

Nav switches to center-admin menus; data scoped to that center

Back to platform exits support view

Center registration: Auth screen → creates center + center admin → empty scoped workspace.

4. The Nine Layers

Layer	Name	Purpose
1	Education CRM	Public site, leads, pipeline, enrollment
2	Academy OS	Batches, schedule, teachers, students, attendance, tests
3	Communication	WhatsApp, email, SMS, automations
4	Student Success	Journeys, homework, interventions, certificates
5	Tutor Success	Lesson plans, homework review, tutor performance, PD
6	Parent Experience	Dashboard, progress, messages, preferences
7	AI Assistants	Rule-based / optional GPT insights
8	Business Intelligence	KPIs, trends, owner decisions
9	Extensions & Partners	Zoom, calendar, templates, partners

5. Core Workflows

5.1 LEAD → STUDENT (LAYER 1)



Pipeline stages: Lead → Inquiry → Demo → Counseling → Admission → Payment → Batch allocation → Student

- Public Website form writes to CRM
- Tabs: Pipeline, All leads, Follow-ups
- Log activities, schedule demos, move stages, enroll to batch
- WhatsApp follow-up from lead detail

5.2 DAILY ACADEMY OPERATIONS (LAYER 2)

Setup once

Add teachers → Create batches → Topics + days/times
→ Auto-generate schedule

Each week

Schedule + links → Mark attendance → Notify parents →
Record tests → Notify parents

- One topic = one session; auto-generate Meet/Zoom/Teams links
- Mark done on completed classes
- Attendance: present / absent / late → optional parent WhatsApp

5.3 HOMEWORK & STUDENT SUCCESS (LAYERS 4 & 5)

Action	Center admin	Teacher	Parent	Student
+ Assignment	✓	✓	—	—
Issue certificate	✓	✓	—	—
+ Feedback	✓	✓	—	—
Submit homework	—	—	mark submitted	submit
View only	—	—	✓	✓

5.4 COMMUNICATION (LAYER 3)

- Channels: WhatsApp, email, SMS, push (simulated in demo)
- Templates & automations: absence, tests, homework due, class reminders
- Comm Hub: broadcasts, settings, message log
- WhatsApp simulate or live via Meta API in Settings

5.5 PARENT EXPERIENCE (LAYER 6)

Tab	Content
Home	Success score, next class, activity, contact teacher
Progress	Strengths, skills, test history
Homework	Assignments and status
Attendance	History and rate
Feedback / Messages / Settings	Teacher notes, thread, notification prefs

5.6 INTELLIGENCE & AI (LAYERS 7 & 8)

Business Intelligence

Decisions feed, KPIs, trends, lead analytics — action cards not static reports

AI Assistants

Role-aware chat; rule-based default; optional OpenAI; drafts parent messages

5.7 EXTENSIONS (LAYER 9)

Connections (Zoom, Calendar, Twilio), ops templates, partner catalog — not a content marketplace.

6. End-to-End Journey (New Center)

Platform owner registers center

Center admin configures Settings; public site + CRM live

Create batches, generate schedule; link teacher accounts

Leads convert to students

Teachers mark attendance, assign homework

Automations notify parents; parent portal active

Owner uses BI decisions and reports

7. Technical Architecture

```
index.html
└─ js/app.js      Auth, navigation, modals
    │ auth.js      Login, register, support view
    │ portals.js   Role nav & access control
    │ store.js     localStorage v5, multi-tenant
    │ views.js     Dashboard, CRM, academy
    │ layer-views.js Layers 3-9 + public site
    │ platform.js  Assignments, parent prefs, etc.
    └─ intelligence.js / ai.js / communication.js
```

DATA MODEL

Entity	Scoped by	Notes
centers[]	platform	Tenant root
users[]	center	Roles, linked IDs
batches , students , leads	centerId	Core ops data
attendance , tests , messages	centerId	Ops records

Session: tutorhub_session · Data: tutorhub_data_v5

RUNNING LOCALLY

Double-click **start.bat** (keep window open)

Open **http://127.0.0.1:8888/**

Do not open file:///.../index.html — ES modules need a local server

8. Demo Accounts

Password for all: **demo123**

Portal	Email
Platform owner	owner@eduos.app
Center admin	admin@brightminds.demo
Teacher	anita@tutorhub.com / vikram@tutorhub.com
Student	aarav@email.com
Parent	rajesh@email.com

Centers: **Bright Minds Academy** (full data) · **Excel Tutors Pune** (empty). Reset: Settings → Reset Demo Data

9. Permission Summary

- **Platform owner** — all centers + support view
- **Center admin** — all layers in center
- **Teacher** — scoped batches/students + homework tools
- **Student** — schedule + homework submit
- **Parent** — view + message + mark homework submitted

10. Out of Scope (MVP)

- Payment collection and invoicing
- Hosting course videos or worksheet libraries
- Production backend / real multi-user sync
- Native mobile apps
- Full API enforcement (demo UI only)

11. Suggested Reading Order

Center admin → Dashboard

CRM → lead to enrollment

Batches → auto-schedule → Class Schedule

Attendance + Tests → Communication Hub

Teacher → Today → attendance → homework

Parent → dashboard → homework

Platform owner → all centers → support view

Document reflects tutor-hub multi-tenant auth release. Update when layers or roles change.